

ASSIGNMENT

Study and implement a basic ethical hacking information-gathering exercise on a deliberately provided/authorized test system. Identify the system's basic information and potential security concerns, document your findings, and explain how an ethical hacker should report and disclose the identified issues responsibly.

Aim

To perform basic information gathering on an authorized test system, identify available network services, analyze possible security concerns, and prepare an ethical disclosure report.

1. Requirements

- Computer/Laptop
- Linux/Windows system
- Terminal/Command Prompt
- Nmap or an equivalent authorized network-scanning tool

2. Ethical Consideration

The test was performed only against the **local system (127.0.0.1)**, which represents the machine on which the test is being performed. No external or unauthorized system was targeted.

3. Implementation

Step 1: Identify the Target

The target selected for the practical was:

Target IP: 127.0.0.1

127.0.0.1 is the loopback address used to refer to the local system.

Step 2: Perform Port Scanning

I performed a basic TCP connection scan against the authorized local system.

Command:

```
nmap -sT -p 22,23,25,53,80,110,135,139,143,443,445,3306,3389,5432,8080 127.0.0.1
```

Output:

PORT	STATE	SERVICE
22/tcp	closed	ssh
23/tcp	closed	telnet
25/tcp	closed	smtp
53/tcp	closed	domain
80/tcp	closed	http
110/tcp	closed	pop3
135/tcp	closed	msrpc
139/tcp	closed	netbios-ssn
143/tcp	closed	imap
443/tcp	closed	https
445/tcp	closed	microsoft-ds
3306/tcp	closed	mysql
3389/tcp	closed	ms-wbt-server
5432/tcp	closed	postgresql
8080/tcp	open	http-proxy

Step 3: Observation

From the scan result, most of the tested ports were found to be closed. However, TCP port 8080 was found to be open and associated with the http-proxy service.

The open port indicates that a service is accepting connections through port 8080. The presence of an open port does not by itself confirm a vulnerability; the service and its configuration need to be reviewed.

Step 4: Security Analysis

The open port 8080 should be checked to determine:

- Which application is using the port.

- Whether the service is required.
- Whether it is properly configured.
- Whether access should be restricted.
- Whether the application is updated.

Then continue with the Ethical Disclosure → Result → Conclusion sections as I gave earlier.

4. Ethical Disclosure

If this finding were discovered during an authorized security assessment, I would report it to the system owner rather than attempting unauthorized access.

Sample Finding

Finding: Open TCP Port 8080

Target: 127.0.0.1

Port: 8080/tcp

Status: Open

Description:

During the authorized port-scanning exercise, TCP port 8080 was found to be open on the local test system. A service is accepting connections through this port.

SecurityConcern:

An unnecessarily exposed service can increase the system's attack surface. The actual risk depends on the application running on the port and its configuration.

Recommendation:

1. Identify the application using port 8080.
2. Verify whether the service is required.
3. Review its configuration.
4. Restrict access if appropriate.
5. Keep the application updated.
6. Disable the service if it is not required.

5. Result

The basic information-gathering exercise was successfully performed on the authorized local system.

The scan identified:

Target : 127.0.0.1

Open Port : 8080/tcp

Service : http-proxy

Other tested ports : Closed

6. Conclusion

Through this implementation, I learned how basic information gathering can identify network services running on a system. The practical also demonstrated the importance of performing security testing only on authorized systems and reporting findings responsibly.

The scan showed that **TCP port 8080 was open**, which should be reviewed by the system administrator to determine the application using the port and whether it is appropriately secured.